

Advanced Simulations: Surveys

Esin G. Gulbahar and Jakob Stierhof on behalf of the SIXTE team

Lea Dauner, Thomas Dauser, Christian Kirsch, Maximilian Lorenz,
Karan Pal, Philipp Thalhammer, Franziska Haluschka, Jörn Wilms

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Outline

We have prepared four sets of simulation scripts that can be run and modified for this hands-on session. The sets can be found in the **surveys** directory in the GitHub repository:

- **chandra_deep_field**

Create a simulation of the Chandra Deep Field South (CDFS) with the NewAthena WFI.

- **dithering**

Create a simulation of the satellite in dithering mode and generate an exposure map of the observation.

- **simple_slew**

Run an all-sky survey simulation for NewAthena WFI for multiple sources.

- **arf_generation**

Run an ARF correction using `sixte_arfgen`.

Where to find the files

Clone the workshop GitHub repository:

https://github.com/stierhofj/Sixte_Workshop_2026.git



simulation_examples / surveys → ..

arf_generation
athena_wfi_cdfs_deep_field
dithering
simple_slew
Exercise sets

input	Extra simulation input files
01_create_simput	Scripts of the chosen exercise folder
02_slew_attitude	
03_simulate	
04_imgev	
05_chip_image	
README.md	Exercise information
chip_image.py	Additional Python scripts
run	Runs all the scripts at once
setup.sh	Setup exercise parameters

Start running the exercises

Feel free to start from any folder; however, do not forget to read the README.md.

Let us know if you have any questions!

athena_wfi_cdfs_deep_field: simulate the Chandra Deep Field South

dithering: simulation of dithering mode and exposure map generation

simple_slew: All-sky survey simulation

arf_generation: ARF correction